

AutoResearch Agent

Demo Script | 8–10 Minutes

Built with Python, Claude API & Streamlit

Before You Start

- Open **http://localhost:8501** in Chrome in fullscreen mode.
- Run: **streamlit run app.py** in your terminal.
- Have your **.env** file ready with your Anthropic API key.
- Start screen recording: **Windows key + Alt + R**.

Demo Steps

- 1 Show the landing screen for a moment without touching anything.**
Say: "This is AutoResearch Agent — an agentic AI that takes any research question, autonomously breaks it down, searches the live web for each angle, and writes you a full cited report. It's built with Python, the Claude API, and Streamlit."
- 2 Point to the hero title and badge at the top.**
Say: "The interface is fully custom styled — dark theme, gradient typography, and a clean layout built in Streamlit with injected CSS."
- 3 Click on the Research Depth slider and move it slowly from 1 to 10 and back to 3.**
Say: "Before running anything, the user controls how thorough they want the research to be. Depth 1 means 1 sub-question for a quick answer. Depth 10 means 10 separate web searches for a comprehensive deep dive. I'll leave it at 3 for this demo."
- 4 Click the Research Question input box. Type: 'What are the long-term effects of social media on teenage mental health?'**
Say: "The agent can handle any open-ended research question — academic, technical, business, or general knowledge."
- 5 Click 'Run Research →'. Point to the first status panel as it expands.**
Say: "Step one is planning. The agent doesn't search blindly — it first uses Claude Haiku to decompose the question into focused sub-questions. Watch them appear one by one."
- 6 Point to each step card as it appears in the planning panel.**
Say: "Each card is a targeted angle the agent will investigate independently. This is what makes it agentic — it plans before it acts, just like a human researcher would."

7	Point to the second status panel as research begins. Say: "Now the real work begins. For each sub-question, the agent runs a live DuckDuckGo search — no API key required, completely free — then feeds those results into Claude to summarize accurately."
8	While researching, point to the progress indicator. Say: "Each sub-question triggers a real web search. The agent reads the top results, extracts the key facts, and moves on to the next angle. Fully autonomous — no human input needed."
9	Point to the third status panel — 'Writing your report...' Say: "Once all the live research is collected, Claude synthesizes everything into a single coherent markdown report — with headers, bullet points, structured sections, and sources."
10	Scroll down slowly to show the full rendered report in the dark card. Say: "The report renders directly in the browser — fully styled with coloured headers, bullet points, and bold text. Every claim in here came from a live web search, not from the model's memory."
11	Scroll to the Sources section at the bottom of the report. Point to the clickable links. Say: "And here's the key part — a full Sources section with clickable links to every page the agent actually read. You can click any of these and verify the information yourself."
12	Click one of the source links to open it in a new tab. Show it briefly then come back. Say: "Live URLs, opening real pages. The agent searched, read, and cited its sources just like a human researcher would."
13	Point to the two download buttons side by side. Say: "The user gets two export options — Markdown for developers and editors, or a styled PDF for sharing with anyone."
14	Click 'Download PDF'. Open it and scroll to the Sources section. Point to the blue underlined links. Say: "Notice the Sources section in the PDF — every URL is rendered as a blue underlined hyperlink. Click any of them and your PDF viewer opens the page directly in the browser. This is a fully professional, shareable document."
15	Go back to the slider. Set depth to 5. Type a new question: 'How does quantum computing threaten modern encryption?'. Click Run Research. Say: "Let me show you depth in action. Five sub-questions, five live DuckDuckGo searches, all on a completely different topic. No configuration changes — just a new question and a higher depth."
16	Let it run. Point to the 5 step cards as they appear. Say: "Five separate angles being researched in real time. The agent adapts its research plan entirely based on the topic — it's not using a fixed template."

17 Once the report appears, scroll to the Sources section.

Say: "Five angles researched, five sets of sources, all synthesized into one report. In production you'd use depth 3 to 5 for most research tasks, and depth 7 or higher for thorough academic-style reports."

18 Stop recording: Windows key + Alt + R.

Say: "To summarise the stack — Python for orchestration, Claude Haiku for planning and synthesis, DuckDuckGo for free live web search, Claude Haiku again to summarize search results, Streamlit for the UI, and ReportLab for PDF generation with clickable citations. The whole pipeline is agentic and produces cited, verifiable reports from live web data. Thanks for watching."

Tips for a Smooth Recording

- **Talk while it runs** — the agent takes 10–20 seconds per question. Use that time to explain what's happening under the hood.
- **Use depth 3 for the first demo** — fast enough to keep momentum, detailed enough to impress.
- **Scroll the report slowly** — give viewers time to read the headers and structure.
- **If Claude gives an unexpected answer** — go with it. It shows the AI is genuinely reasoning, not just templating.
- **Mention token efficiency** — the app uses Haiku, the lightest Claude model, keeping costs extremely low.
- **Keep water nearby** — 8 minutes of talking is longer than it sounds.